Class Notes: Deep Learning & Neural Networks

1. Supervised Learning: Models are trained on labeled pairs (x, y).

   Examples: Linear Regression, Decision Trees, CNNs for Vision.

2. Unsupervised Learning: Clustering hidden representations without labels.

   Examples: K-Means Clustering, Principal Component Analysis (PCA).

3. Overfitting Prevention: Use Dropout layers and L2 Regularization.